Safety and Efficacy of Seqirus A/H7N9 Inactivated Influenza Vaccine With or Without MF59® Adjuvant to Prevent Avian Influenza (H7N9)

NCT Identifier: NCT03682120
Sponsor: National Institute of Allergy and Infectious Diseases (NIAID)
IND Sponsor / Manufacturer: Seqirus Inc. (formerly bioCSL / Novartis Vaccines)
Phase: Phase 2
Study Type: Interventional — Randomized, Observer-Blinded
Protocol Version: Version 3.0
Date: 10 September 2018
Registry Status: Publicly registered at ClinicalTrials.gov

This document is derived from publicly available information registered on ClinicalTrials.gov under NCT03682120.

1. Background and Scientific Rationale

Influenza A(H7N9) viruses first emerged in eastern China in February 2013, causing severe human respiratory illness with high mortality (case fatality rate >30%). Through 2018, the WHO reported five epidemic waves totaling more than 1,560 laboratory-confirmed human cases. Although sustained human-to-human transmission has not been documented, H7N9 viruses have acquired mammalian-adapting mutations and are classified by the CDC/WHO as viruses with pandemic potential.

NIAID has supported development of candidate H7N9 inactivated influenza vaccines (IIV). MF59® (Seqirus) is a squalene oil-in-water emulsion adjuvant that enhances hemagglutination inhibition (HAI) antibody responses to inactivated influenza vaccines, potentially enabling antigen dose-sparing and broader cross-reactive responses—both critical for pandemic preparedness.

2. Study Objectives

2.1 Primary Objective

Evaluate the safety and tolerability of two intramuscular doses of Seqirus A/H7N9 IIV administered with or without MF59® adjuvant in healthy adults aged 19–64 years.

2.2 Secondary Objectives

- Evaluate immunogenicity: HAI and microneutralization (MN) titers at Days 0, 29, 57, and 209.
- Determine seroconversion rate (≥4-fold HAI titer rise from baseline) after the 2-dose series.
- Compare immune responses between adjuvanted and non-adjuvanted formulations.
- Assess antigen dose-sparing potential of MF59® adjuvant.
- Evaluate 6-month durability of antibody responses.

3. Study Design

Phase 2, randomized, observer-blinded, controlled, multi-center trial at NIAID-supported Vaccine and Treatment Evaluation Units (VTEUs). Participants randomized to one of four dose/adjuvant groups (block randomization, stratified by site). All participants receive two IM injections 28 days apart (Days 0, 29).

Group	Vaccine	Adjuvant	HA Antigen Dose	N
A	Seqirus A/H7N9 IIV	MF59®	3.75 µg	50
B	Seqirus A/H7N9 IIV	MF59®	7.5 µg	50
C	Seqirus A/H7N9 IIV	None	15 µg	50
D	Seqirus A/H7N9 IIV	None	45 µg	50

4. Eligibility Criteria

4.1 Inclusion Criteria

- Healthy adults aged 19–64 years at enrollment.
- Able to attend all study visits and comply with protocol requirements.
- Negative serum pregnancy test (women of childbearing potential); willing to use effective contraception for 60 days post-last vaccination.
- No prior H7N9 vaccination or laboratory-confirmed H7N9 infection.
- Negative baseline HAI titer to A/H7N9 (≤1:10).

4.2 Exclusion Criteria

- Immunosuppressive therapy or immunocompromising condition within 6 months.
- Receipt of live vaccine within 30 days or inactivated vaccine within 14 days of enrollment.
- Known allergy to eggs, egg products, or any vaccine component including squalene.
- Pregnancy or breastfeeding.
- History of Guillain-Barré syndrome within 6 weeks of a prior influenza vaccination.
- Significant cardiovascular, pulmonary, hepatic, or renal disease.
- Current participation in another interventional clinical study.

5. Vaccination Schedule and Procedures

Both vaccinations are delivered by IM injection into the non-dominant deltoid on Days 0 and 29. Participants are observed for ≥30 minutes post-vaccination. A 7-day electronic diary (eDiary) captures solicited local and systemic reactions after each injection.

Visit	Day	Key Procedures
Screening	–28 to –1	Consent, medical history, physical exam, labs, serology (HAI baseline)
Vaccination 1	0	Vaccination, safety labs, serum (HAI/MN baseline)
Follow-up 1	7	Safety assessment
Vaccination 2	29	Vaccination, serum (immunogenicity)
Follow-up 2	36	Safety assessment
Immunogenicity	57	Serum (HAI/MN, primary immunogenicity)
Durability	209	Serum (HAI/MN long-term), final safety

6. Outcome Measures

6.1 Safety Outcomes

- Solicited local reactions (pain, redness, swelling) and systemic reactions (fever, fatigue, headache, myalgia) for 7 days post-each vaccination.
- Unsolicited adverse events for 28 days post-each vaccination.
- Medically attended adverse events (MAAEs) through Day 209.
- Serious adverse events (SAEs) through Day 209.

6.2 Immunogenicity Outcomes

- HAI antibody titers at Days 0, 29, 57, 209.
- Microneutralization (MN) antibody titers at Days 0, 29, 57, 209.
- Geometric mean titer (GMT) and GMT ratios comparing adjuvanted vs. non-adjuvanted groups.
- Seroconversion rate: proportion with ≥4-fold rise from baseline HAI titer by Day 57.
- Seroprotection rate: proportion with HAI titer ≥1:40 by Day 57.

7. Statistical Analysis Plan

7.1 Sample Size

50 evaluable subjects per group (200 total) provides 80% power to detect a 15-percentage-point difference in seroconversion rates at two-sided $\alpha = 0.05$, assuming 50% baseline seroconversion. Enrollment target of 55 per group accommodates ~10% attrition.

7.2 Immunogenicity Analysis

GMT ratios (95% CIs) calculated via ANCOVA on log-transformed titers, adjusting for baseline titer and site. Seroconversion and seroprotection rates compared between groups using Fisher's exact test with Bonferroni correction.

8. Regulatory Considerations

Conducted under a NIAID DMID master IND held with the US FDA. SAEs are reported to NIAID, the IND sponsor, and institutional IRBs within 24 hours of site awareness. All participating sites are approved under the NIAID DMID VTEU master agreement.